

GMP validation — all nine transformers

AIMMS 26.3.2.1, CPLEX 22.1, thc22-PC · runs 10–13 August 2026 · rolling-horizon FOR, 48 half-hour periods × 12 directions

Runtime with GMP. The two clocks measure different things:

- **Solver** — the model's own time column, which is `FOR_VertexCnt.SolutionTime` summed over the 576 sweep vertices. It is the time CPLEX spends inside those 576 optimizations, and nothing else.
- **Wall-clock** — elapsed real time from run start to run finish. It contains the solver time *plus* everything the solver column cannot see: reading the network and load data, generating the matrix, the 48 committed baseline solves, and writing the CSVs.

Wall-clock is therefore the honest measure of what a run costs, and the only one that can judge GMP: GMP works by removing matrix *generation*, which the solver column never records. Across the nine runs the gap is 35 h 43 m – 30 h 18 m = **5 h 25 m**, so about **15 %** of a run happens outside the vertex solves.

Transformer	Batteries	Wall-clock	Solver	Wall/solver	s / solve
TR4	46	0 h 36 m	0 h 31 m	1.16	3.19
TR3	49	0 h 39 m	0 h 34 m	1.15	3.53
TR1	77	1 h 37 m	1 h 23 m	1.16	8.68
TR2	95	2 h 20 m	2 h 02 m	1.15	12.73
TR9	90	3 h 16 m	2 h 50 m	1.15	17.76
TR8	124	4 h 59 m	4 h 15 m	1.17	26.65
TR6	262	6 h 49 m	5 h 34 m	1.22	34.84
TR7	259	6 h 51 m	5 h 46 m	1.19	36.06
TR5	214	8 h 36 m	7 h 21 m	1.17	45.98
Total	1216	35 h 43 m	30 h 18 m	1.18	

Rows are ordered by cost. The five largest runs are **89 %** of the total. Cost does not track battery count — TR5 with 214 batteries is the most expensive run of the nine, above TR7 (259) and TR6 (262). Topology drives it, not prosumer count.

Wall-clock came from file timestamps: the sweep opens its CSV at the start of the run and finishes writing at the end, so creation and last-write bracket it. Validated against three runs whose start times were known, agreeing within 1–2 minutes.

Is GMP worth enabling? Yes — 3.5× faster. Measured by running plain `RunTR2` and `RunTR2_GMP` on the same build and machine, over the same 24 periods (288 vertices).

TR2, same model	AIMMS	s / solve
without GMP	26.1 / 26.2 (July)	10.76
without GMP	26.3.2.1	47.36
with GMP	26.3.2.1	13.61

AIMMS 26.3.2.1 made the non-GMP path about 4.4× slower. GMP insulates against that and lands near the old cost. It is faster in every individual period, not just on average. **On 26.3.2.1, use the _GMP procedures.**

Comparing the July reference times against the new runs would suggest the opposite. That comparison is invalid: the July runs used an older AIMMS, so the difference measures the version, not GMP. The July timings are unreliable in any case — TR9 was run twice before any performance work, giving bit-identical results 2.28× apart in speed.

Correctness: the performance work changed nothing.

	TR4	TR3	TR1	TR9	TR2	TR8	TR5	TR7	TR6
max Δproj (p.u.)	1e-6	1e-6	1e-6	3e-6	1e-6	1e-6	1e-6	8e-6	2e-6
committed dispatch	identical — max d = 0.000e+00 on every row, every column								

5 184 FOR vertices, all Optimal in both runs, no status change anywhere. **58 368 dispatch rows** across 1 216 batteries, every one identical — not within tolerance, identical. The worst boundary movement in the campaign is 8e-6 p.u. = **8 W** at $S_b = 1$ MW, against the model's own feasibility tolerance of 1 kW.

The published analysis is unaffected. The nine `TRn_analysis_data.xlsx` workbooks were rebuilt from the new dispatch — **about 234 000 cells**, worst deviation 8.5e-14, float64 rounding on the unit conversions. All **72 published headline metrics** (PV, demand, export, import, SCR, SSR, periods within 50 W, simultaneous charge/discharge rows) reproduce for all nine, including TR9's two documented charge/discharge overlap rows and the zero on the other eight.

Full evidence, per-transformer reports and regeneration scripts: `Tese_Organization_July22_26\GMP_VALIDATION_2026-08\`. The `RunTR2` reference was still running at the time of writing — period 25 of 48 after 5 h 27 m, against the GMP run's 2 h 20 m total.